Design of a Step-Down (Buck) DC-DC Converter

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Dataset **D**

1 Introduction

This project aims to address the design task of a step-down (buck) DC-DC converter that can efficiently regulate voltage by the reduction of the input voltage to a specific lower voltage level in the output, based on given specifications. The importance of solving this problem emerges since converters are widely used in energy systems, electric vehicle chargers, and industrial DC supply units, where stable power conversion under dynamic conditions is crucial for system reliability and efficiency. The requirements for the converter's design, such as input and output voltage range, load current, and allowable output voltage ripple, are provided in the Dataset-D in Table 4.

In this assignment, the choice of the right semiconductor devices, the calculation of the switching frequency, the losses of our power devices, and the choice of the heat sink are crucial to guarantee the efficient and the safe operation of the system. Moreover, the circuit simulator of PSpice, is used for analyzing the final design by means of simulations using 3 scenarios: Open Loop simulation in Section 3.1, Closed-Loop Control simulation in Section 3.2 and Load Step simulation in Section 3.3.

2 Dimensioning of the Circuit

The buck converter must be designed to operate in Continuous Conduction Mode (CCM) across the entire input voltage range. Based on the specifications provided in the dataset and the available semiconductor components, the appropriate values are selected in this section. Moreover, switching frequency is determined. Thermal considerations are also taken into account. Since the active devices (IGBT and diode) are subject to both conduction and switching losses, they must be mounted on a heat sink to ensure their junction temperatures remain within safe operating limits. Therefore, the thermal design, including heat sink dimensioning, is a part of ensuring reliable and efficient operation of the converter.

2.1 Selection of Diode and IGBT

The selection of active components was based on the dataset requirements and the pool of available devices, with the aim of ensuring safe, reliable, efficient, and cost-effective operation of the converter. According to the Dataset requirements in Table 4, the input voltage ranges from 500 V to 580 V, and the maximum load current is 20 A.

Given the possibility of voltage overshoots due to switching transients, a suitable voltage margin must be applied. For this reason, only devices rated at 1200 V are considered appropriate for both the switch (IGBT) and the diode.

Among the IGBT candidates, the **APT20GF120BR** was selected. It offers a voltage rating of 1200 V and supports a continuous current of 20 A at 90°C, precisely matching the design requirement. Although other options such as the APT50GF120JRD provide higher current ratings and improved thermal characteristics, they are unnecessarily oversized for this application and less cost-effective.

For the diode, the **APT30D120B** was chosen. It is also rated at 1200 V and can withstand a continuous current of 30 A, which provides a safe operating margin above the 20 A load current. Other diodes options such as the APT60D120B were evaluated, but dismissed for the reason of overdimensioning and cost efficiency.

In summary, the selected components meet all voltage and current constraints while avoiding unnecessary cost, size, and thermal complexity. This ensures the converter operates within its safe limits and aligns with the technical and economic objectives. The documents of selected components are attached in the section of the Appendix.

2.2 Switching Frequency

To ensure proper operation in Continuous Conduction Mode (CCM), an appropriate switching frequency f_{sw} must be selected. The frequency should satisfy the constraint defined by the maximum permissible product of inductance and capacitance, i.e., $L \cdot C < 9 \cdot 10^{-8}\,\mathrm{s}^2$, and the allowable output voltage ripple ΔV_o.

The duty cycle D is given by

$$D = \frac{V_o}{V_{\text{in}}}, \tag{1}$$

where V_{in} and V_o are the input and output voltages of the converter, respectively. For the worst-case scenario, we use $V_{\text{in}} = 500\,\text{V}$ (minimum value) and $V_o = 260\,\text{V}$, leading to $D = 0.52$.

According to Eq. (7-23) from [1], the minimum switching frequency that meets the ripple constraint is calculated as:

$$f_{\text{sw}} = \sqrt{\frac{V_o(1-D)}{8LC\Delta V_o}}, \tag{2}$$

where $L \cdot C = 9 \cdot 10^{-8}\,\text{s}^2$ and $\Delta V_o = 4\,\text{V}$. Substituting the values, we obtain:

$$f_{\text{sw}} = \sqrt{\frac{260 \cdot (1-0.52)}{8 \cdot 9 \cdot 10^{-8} \cdot 4}} \approx 6581.5\,\text{Hz}.$$

Rounding the result to the nearest multiple of 1 kHz, the switching frequency is chosen as:

$$f_{\text{sw}} = 7\,\text{kHz}. \tag{3}$$

This value ensures compliance with the ripple constraint and the $L \cdot C$ product limitation, and will be used in all subsequent calculations for power loss and thermal analysis.

2.3 Losses in Power Devices

The selected semiconductor devices that were selected in Section 2.1 exhibit two main types of power losses: conduction or on-state losses (P_{on}) and switching losses (P_{sw}). These losses can cause a significant rise in the device's operating temperature and must be effectively dissipated in order to avoid thermal damage and ensure reliable and safe system operation.

2.3.1 On-state Losses

All power semiconductor devices experience conduction losses during their on-state operation. These losses are primarily determined by the on-state voltage drop across the device and the current flowing through it. The magnitude of the loss depends on the specific operating conditions, such as current amplitude and junction temperature, and can be estimated using characteristic curves from the device datasheet. The general expression for the average on-state power loss over one switching period is:

$$P_{\text{on}} = \frac{1}{T_{\text{SW}}} \int_0^{T_{\text{SW}}} v_{\text{on}} \cdot i_T \, dt \tag{4}$$

For the IGBT, the on-state power loss is calculated using the collector current I_C and the corresponding collector-to-emitter saturation voltage V_{CE}, while for the diode, the forward current I_F and forward voltage drop V_F are considered. According to specifications, the maximum load current is 20 A, and it is assuming a junction temperature of 150°C, which represents the worst-case thermal condition.The duty cycle D is estimated by Eq. (1)

From the provided datasheet of the selected IGBT (**APT20GF120BR**) (in Figure 3 of the IGBT's document), the typical value of V_{CE} at $I_C = 20\,\text{A}$ and $T_J = 150°\text{C}$ is approximately 3.6 V. So, the on-state conduction loss of IGBT is:

$$\begin{aligned} P_{\text{on, IGBT}} &= D \cdot V_{CE} \cdot I_C \\ &= 0.52 \cdot 3.6 \cdot 20 = 37.44\,\text{W} \end{aligned} \tag{5}$$

For the selected diode (**APT30D120B**), the forward voltage drop V_F at a forward current of $I_F = 20\,\text{A}$ and junction temperature of 150°C can be approximated from the figure 2 of the datasheet as 1.45 V. Since the diode conducts during the off-time of the switch, the average conduction loss is scaled by $(1-D)$, where D is the duty cycle by Eq. (1). Therefore,on-state conduction loss for the diode is:

$$\begin{aligned} P_{\text{on, diode}} &= (1-D) \cdot V_F \cdot I_F \\ &= (1-0.52) \cdot 1.45 \cdot 20 = 13.92\,\text{W} \end{aligned} \tag{6}$$

2.3.2 Switching Losses

The switching losses in power semiconductors depend on the switching frequency and the instantaneous current and voltage at which the device is turned on and off. In this analysis, switching losses are considered only for the IGBT, as the reverse recovery losses of the freewheeling diode are neglected. The IGBT datasheet provides typical values for the turn-on and turn-off energy losses, $E_{\text{on,ref}}$ and $E_{\text{off,ref}}$, at specific reference conditions. These values must be scaled to the actual operating conditions of the converter using the following equation:

$$E_{\text{on/off}} = E_{\text{on/off,ref}} \cdot \frac{v_{\text{ref}} \cdot i_{\text{ref}}}{v_T \cdot i_T} \tag{7}$$

where v_T and i_T are the actual voltage and current during switching, and v_{ref}, i_{ref} are the datasheet reference values under which $E_{\text{on,ref}}$ and $E_{\text{off,ref}}$ were measured.

According to the datasheet of the proposed IGBT (Figure 13), it is observable that at $I_C = 20\,\mathrm{A}$ and $V_{CC} = 0.66 \cdot V_{\mathrm{CES}} = 792\,\mathrm{V}$, with a junction temperature $T_J = 125°\mathrm{C}$, the typical energy values are $E_{\mathrm{on,ref}} \approx 1.2\,\mathrm{mJ}$ and $E_{\mathrm{off,ref}} \approx 1.3\,\mathrm{mJ}$.

By taking into account the values used in Dataset D and applying Equation (7), the scaled switching energies are calculated as:

$$E_{\mathrm{on}} = 0.879\,\mathrm{mJ} \tag{8}$$

$$E_{\mathrm{off}} = 0.952\,\mathrm{mJ} \tag{9}$$

The total IGBT switching losses are calculated:

$$P_{\mathrm{sw,\,IGBT}} = \frac{E_{\mathrm{on}} + E_{\mathrm{off}}}{T_{\mathrm{sw}}} \tag{10}$$

By substituting into Eq. (10) the values obtained from Equations (3), (8), (9), the total switching losses of the IGBT are calculated as:

$$P_{\mathrm{sw,\,IGBT}} \approx 12.82\,\mathrm{W} \tag{11}$$

2.3.3 Total Losses

The total power loss in the IGBT is given by the sum of its conduction and switching losses. By substituting the values from Eq. (5) and Eq. (11), it is obtained:

$$P_{\mathrm{total,\,IGBT}} = P_{\mathrm{on,\,IGBT}} + P_{\mathrm{sw,\,IGBT}} = 50.26\,\mathrm{W} \tag{12}$$

Finally, for the diode, the total loss is equal to the conduction loss only:

$$P_{\mathrm{total,\,diode}} = P_{\mathrm{on,\,diode}} = 13.92\,\mathrm{W} \tag{13}$$

Table 1: Summary of power losses

Device	P_{on} (W)	P_{sw} (W)	P_{total} (W)
IGBT	37.44	12.82	50.26
Diode	13.92	–	13.92

2.4 Heatsink Design

To ensure safe thermal operation of the power converter, an appropriate heat sink must be selected to dissipate the total power losses generated in the IGBT and the diode. The thermal equivalent circuit is shown in Figure 1, it includes all the relevant thermal resistances and it derived through the project description document.

[redacted]

Figure 1: Equivalent thermal network.

Based on the calculated total power losses for each device from Table. 1 and the thermal resistances from junction to case, $R_{\mathrm{th,jc,IGBT}}$ and $R_{\mathrm{th,jc,Diode}}$, obtained from the corresponding datasheets, the maximum allowable heat sink temperatures can be determined.

The maximum allowable junction temperature for both the IGBT and the diode is specified as $T_j = 150°\mathrm{C}$. However, to ensure safe operation and provide an additional thermal margin, a reduced allowable temperature of $T_j = 145°\mathrm{C}$ is assumed. Furthermore, as stated in the project instructions, the case-to-sink thermal resistance is neglected. Under these assumptions, the general expression used to calculate the maximum heat sink temperature for each device becomes:

$$T_{s,\max} = T_j - P_{\mathrm{loss}} \cdot R_{\mathrm{th,jc}} \tag{14}$$

By Applying (14) to each device:

$$T_{s,\max,\mathrm{IGBT}} = 145°\mathrm{C} - 50.26\,\mathrm{W} \cdot 0.63\,°\mathrm{C/W} = 113.33°\mathrm{C} \tag{15}$$

$$T_{s,\max,\mathrm{Diode}} = 145°\mathrm{C} - 13.92\,\mathrm{W} \cdot 0.90\,°\mathrm{C/W} = 132.48°\mathrm{C} \tag{16}$$

Based on the results, the most restrictive case corresponds to the IGBT, which yields a maximum allowable heat sink temperature of $T_{s,\max} = 113.33°\mathrm{C}$ and to ensure a safer margin of operation, the final maximum allowable heat sink temperature is conservatively set to $T_{s,\max} = 110°\mathrm{C}$. The heat sink must be able to cool the system in this defined temperature. The parameters are summarized in Table 2.

Given the ambient temperature $T_a = 50°\mathrm{C}$, the required thermal resistance from sink to ambient, $R_{\mathrm{th,sa}}$, can be calculated using the following expression:

$$R_{\mathrm{th,sa}} = \frac{T_{s,\max} - T_a}{P_{\mathrm{total,IGBT}} + P_{\mathrm{total,Diode}}} \tag{17}$$

$$R_{\mathrm{th,sa}} = \frac{110 - 50}{64.18} \approx 0.93\,°\mathrm{C/W}$$

Table 2: Summary of thermal network parameters.

Parameter	Value	Unit
P_{IGBT}	50.26	W
P_{diode}	13.92	W
$R_{\mathrm{th,jc,IGBT}}$	0.63	°C/W
$R_{\mathrm{th,jc,diode}}$	0.90	°C/W
$T_{j,\max}$	145	°C

To estimate the geometry of the heat sink required for proper thermal management. Both radiative and convective thermal resistances must taken into account. The goal is to ensure that the total thermal resistance from the heat sink to the ambient does not exceed the previously calculated value of $R_{\mathrm{th,sa}} = 0.93\,°\mathrm{C/W}$. The methodology described in [1] (Chapter 29, Section 29-4-3) is followed.

The thermal resistance due to radiation is estimated by:

$$R_{\theta,\mathrm{rad}} = \frac{\Delta T}{5.1 A_{\mathrm{rad}} \left[\left(\frac{T_s}{100}\right)^4 - \left(\frac{T_a}{100}\right)^4 \right]} \tag{18}$$

The required radiating area is approximated as:

$$A_{\mathrm{rad}} = 2 d_{\mathrm{vert}} \cdot (\text{length} + \text{width}) \tag{19}$$

The convective thermal resistance is calculated by:

$$R_{\theta,\mathrm{conv}} = \frac{1}{1.34 A F_{\mathrm{red}}} \left(\frac{d_{\mathrm{vert}}}{\Delta T} \right)^{1/4} \tag{20}$$

The convection area is calculated by (assuming 11 fins):

$$A = 2A_2 + 22A_1 \tag{21}$$

Moreover, F_{red} is a reduction factor that accounts for fin spacing effects, typically found from graphs or approximated (e.g., $F_{\mathrm{red}} = 0.78$ for 9 mm fin spacing as discussed in Mohan).

The total sink-to-ambient thermal resistance $R_{\theta,\mathrm{sa}}$ is determined by combining the radiative and convective components in parallel:

$$R_{\theta,\mathrm{sa}} = \frac{R_{\theta,\mathrm{rad}} \cdot R_{\theta,\mathrm{conv}}}{R_{\theta,\mathrm{rad}} + R_{\theta,\mathrm{conv}}} \tag{22}$$

After the calculations $R_{\theta,\mathrm{sa}} = 0.7\,°\mathrm{C/W}$ which is acceptable for this application. The geometry of the heat sink is observable in Figure 2 [1].

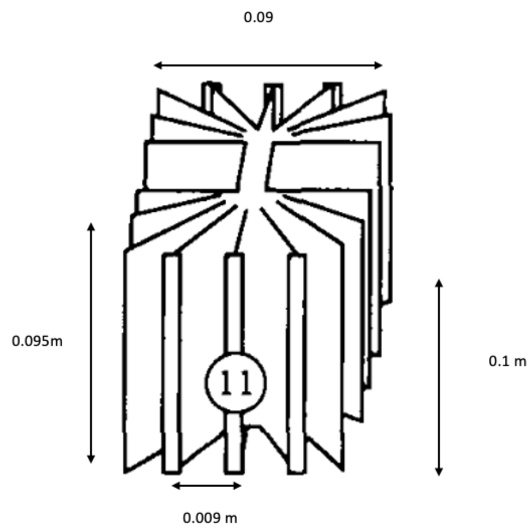


Figure 2: Geometry of the heat sink.

3 Simulation and Control

In the next sections three different simulations are presented: open-loop control, closed-loop control and load step. A crucial part is the calculations of the passive components and to determine the right parameters like the duty ratio according to input-output voltage relationship of and the values of the internal circuit of the buck converter to satisfy the neccessary conditions in each case. All the simulations are performed with the simulation package of PSpice, the results and the waveforms of each case will be discussed.

3.1 Open-Loop Control

The purpose of this simulation is to verify the correct operation of the converter under open-loop control, ensuring that the output voltage and ripple comply with the specifications provided in Table 4. In open-loop mode, the output voltage is not fed back to the control system. Therefore, the passive components L and C must be dimensioned appropriately to satisfy the output ripple and current ripple specifications.

According to Dataset D, the nominal output voltage is $V_o = 260\,\mathrm{V}$, with a maximum output current of $I_{o,\max} = 20\,\mathrm{A}$. The allowable output voltage ripple is $\Delta v_o = 4\,\mathrm{V}$. The switching frequency and duty cycle are selected such that the output voltage meets the target of 260 V. The input voltage is set to $V_{\mathrm{in}} = 500\,\mathrm{V}$ for this simulation.

To estimate the passive components, the following standard formulas are employed for the inductor current ripple and the output voltage ripple

[1]. This methodology for the values of the components was obtained to give a more realistic approach rather than the randomly picked values of the passive components:

$$L = \frac{V_o}{\Delta I_L f_{sw}}(1 - D) \tag{23}$$

$$C = \frac{1}{8} \cdot \frac{\Delta I_L}{\Delta V_o \cdot f_{sw}} \tag{24}$$

For the estimation the next assumption was used. as a fluctuation of 15% to 20% in the current ripple could be considered as rational Thus:

$$\Delta I_L = 0.2 \cdot I_{o,\max} = 0.2 \cdot 20\,\mathrm{A} = 4\,\mathrm{A}$$

Solving (23) and (24):

$$L \approx 4.46\,\mathrm{mH}$$

$$C \approx 17.9\,\mu\mathrm{F}$$

The specification requires $C \cdot L < 9 \cdot 10^{-8}\, s^2$, which is satisfied.

Finally, the internal resistance of the inductor and the equivalent series resistance (ESR) of the capacitor are calculated according to the specifications provided in Table 4:

$$r_L = K_L \cdot L^{0.7} = 7.97 \cdot (4.46 \times 10^{-3})^{0.7} \approx 0.180\,\Omega$$

$$r_C = \frac{K_{\mathrm{ESR}}}{C} = \frac{8.5\,\mu\Omega \cdot \mathrm{F}}{17.9\,\mu\mathrm{F}} \approx 0.475\,\Omega$$

The load resistance is selected to ensure $I_o = 20\,\mathrm{A}$ at $V_o = 260\,\mathrm{V}$:

$$R_{\mathrm{load}} = \frac{V_o}{I_{o,\max}} = \frac{260}{20} = 13\,\Omega.$$

The PWM parameters are set as:

$$\text{Period} = \frac{1}{f_{sw}} = \frac{1}{7000} \approx 143\,\mu\mathrm{s}$$

$$\text{Pulse Width} = D \cdot \text{Period} = 0.52 \cdot 143\,\mu\mathrm{s} \approx 74.36\,\mu\mathrm{s}$$

With these component values, the circuit is prepared for open-loop simulation in PSpice. The results regarding inductor current ripple and output voltage ripple will be compared to the theoretical values. The results are presented in Figures 3, 4, and 5.

After the transient period, the output voltage and inductor current stabilize around their desired values.

Using the simulation cursor, the peak-to-peak current ripple of the inductor is measured as:

$$I_{L,\max} = 22.862\,\mathrm{A} \quad \text{and} \quad I_{L,\min} = 18.899\,\mathrm{A}$$

Table 3: Summary of component and value parameters.

Parameter	Value
Input Voltage	500 V
Output Voltage	260 V
Inductance L	4.46 mH
Capacitance C	17.9 μF
Switching Frequency f_{sw}	7 kHz
Load Resistance R_{load}	13 Ω
Duty Cycle D	0.52
Inductor Resistance r_L	0.180 Ω
Capacitor ESR r_C	0.475 Ω
IGBT	APT20GF120BR
Diode	APT30D120B

$$\Delta I_L = 22.862 - 18.899 = 3.963\,\mathrm{A}$$

Similarly, for the output voltage:

$$V_{o,\max} = 269.423\,\mathrm{V} \quad \text{and} \quad V_{o,\min} = 265.509\,\mathrm{V}$$

$$\Delta V_o = 269.423 - 265.509 = 3.914\,\mathrm{V}$$

Both ripple values are close to the design specifications (4 V for voltage ripple and approximately 4 A for current ripple), confirming the validity of the passive component selection.

Figure 3: Output voltage and inductor current during open-loop simulation

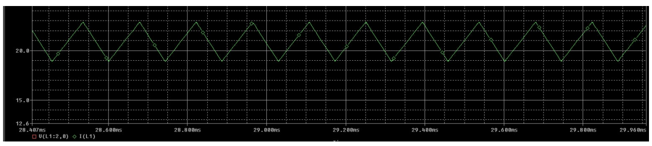


Figure 4: Inductor current ripple during open-loop simulation

Figure 5: Output voltage rippleduring open-loop simulation

3.2 Closed-Loop Control

Variation in the input voltage and other disturbances may occur. A negative feedback loop is employed to regulate the output voltage as close as possible to the reference value.

3.2.1 Power Stage, Output Filter and PWM Controller

The transfer function of the power stage and the output filter is given by:

$$T_p(s) = V_d \cdot \frac{\omega_0^2}{\omega_z} \cdot \frac{s + \omega_z}{s^2 + 2\xi\omega_0 s + \omega_0^2} \tag{25}$$

where:

$$\omega_0 = \frac{1}{\sqrt{LC}} \qquad \omega_z = \frac{1}{\text{ESR} \cdot C} \tag{26}$$

and

$$\xi = \frac{\dfrac{1}{C \cdot R_{\text{load}}} + \dfrac{\text{ESR} + R_L}{L}}{2\omega_0} \tag{27}$$

The transfer function of the PWM modulator is expressed as:

$$T_m(s) = \frac{1}{\hat{V}_R} \tag{28}$$

where $\hat{V}_R$ is the peak value of the carrier waveform.

Thus, the total open-loop transfer function from v_c to v_o becomes:

$$T_1(s) = T_p(s) \cdot T_m(s) \tag{29}$$

3.2.2 Compensated Error Amplifier and Total Transfer Function

The total open-loop transfer function is expressed as:

$$T_{\text{OL}}(s) = T_1(s) \cdot T_c(s) \tag{30}$$

where $T_c(s)$ represents the transfer function of the compensated error amplifier. A suitable starting point for the compensator is the following structure:

$$T_c(s) = \frac{A}{s} \cdot \frac{s + \omega_{z2}}{s + \omega_p} \tag{31}$$

3.2.3 Design of the control circuit

The design is based on the following procedure:

- The crossover frequency was selected as $\omega_{\text{cross}} = 2\pi \cdot \frac{f_{sw}}{10} = 4398.2$ rad/s, with switching frequency $f_{sw} = 7$ kHz.

- Using MATLAB, the open-loop transfer function $T_1(s)$ was evaluated at ω_{cross}, yielding a gain of -4.36 dB and a phase of $-107.09°$.

- A phase margin of 60° was chosen. Assuming $\phi_m = 0°$, the required phase of the compensator at crossover is:

$$\begin{aligned}\phi_c &= -(180° - PM - \phi_m - \phi_p) \\ &= (180° - 60° - 0° - (-107.09°)) = -12.91°\end{aligned} \tag{32}$$

- The phase boost is computed as:

$$\text{boost} = 90° + \phi_c = 90° - 12.91° = 77.09° \tag{33}$$

- The K-factor is calculated as:

$$K = \tan\left(45° + \frac{\text{boost}}{2}\right) = 8.84 \tag{34}$$

- The compensator's frequencies are given by:

$$\omega_{z2} = \frac{\omega_{\text{cross}}}{K} = 497.66 \text{ rad/s} \tag{35}$$

$$\omega_p = K \cdot \omega_{\text{cross}} = 38871.06 \text{ rad/s} \tag{36}$$

- The gain A from Equation 31 is chosen so that the total gain at crossover is 1 or 0 dB. The resulting A is found by using the values from Equations 35 and 36 and it is 64198.36.

With these values, the circuit is prepared for closed-loop simulation in PSpice.

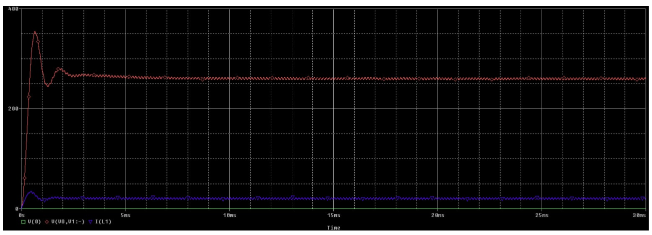


Figure 6: Output voltage and inductor current during closed-loop simulation

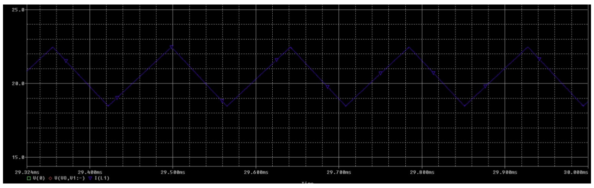


Figure 7: Inductor current ripple during closed-loop simulation

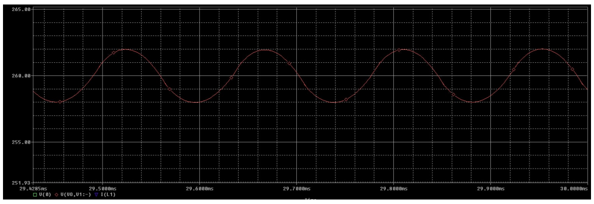


Figure 8: Output voltage ripple during closed-loop simulation

By comparing the open-loop figures 3, 4 and 5 with closed-loop figures6, 7 and 8, it is evident that the closed-loop control achieves better performance towards the reference output voltage of 260V. Both of them achieve sufficient performance in the aspect of voltage and current ripple. Moreover, it is observable that the closed-loop system exhibits an initial overshoot and small oscillations during the transient period before settling around the desired output voltage. In general this specific type of control action ensures better regulation of the buck converter based on the requirements.

3.3 Load Step

During the load-step test, the load current is increased from 15 A to 20 A by connecting an additional resistor in parallel with the existing load. This instantaneous change in the output current demand causes disturbances in both the output voltage and the inductor current, as respectively presented in Figures 9 and 10.From the simulation results, it is observed that immediately after the load step at t = 15 ms, there is a clear transient behavior on the output voltage. More specifically, There is a sudden voltage dip but after a while the control system reacts gradually recovers and stabilizes back to the reference voltage of approximately 260 V, with small oscillations fading over time. It is also observable that the current increases in step with the load change and stabilizes at the new higher level, as expected. The selected values of L and C contribute to relatively smooth inductor current behavior but introduce a small transient in voltage recovery, along with some oscillations before settling. Nevertheless, the system successfully adjusts and regulates the output voltage to the desired value after the disturbance.

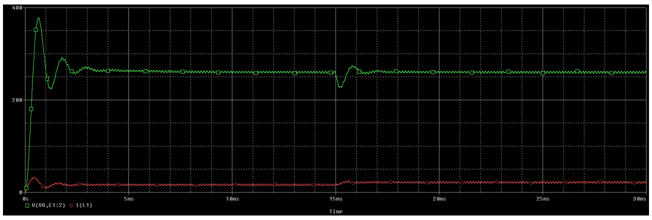


Figure 9: Output voltage and inductor current during step-load simulation

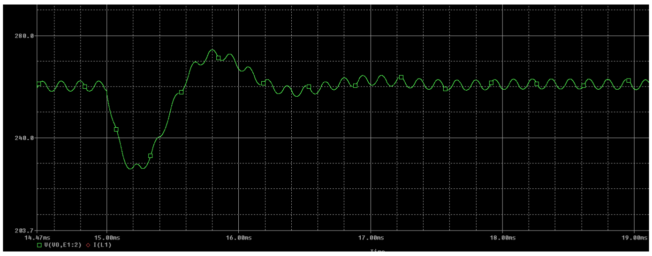


Figure 10: Output voltage ripple during step-load simulation

4 Conclusion

This project provided a comprehensive study and implementation of a step-down (buck) DC-DC converter design based on the given specifications. The key objective was to ensure reliable and efficient voltage conversion while satisfying requirements related to output voltage, load current, and ripple limits. The design procedure covered the careful selection of semiconductor devices, passive components, switching frequency, and thermal management considerations.

In general, the simulation results validate the correctness of the design choices. Both open-loop and closed-loop simulations confirmed that this design of the converter can achieve the desired output voltage of 260 V with acceptable ripple levels in both voltage and current, in line with the given specifications. Specifically, as it was expected the closed-loop simulation demonstrated improved voltage regulation performance over the open-loop design. Moreover, the load-step analysis confirmed the converter's capacity to handle dynamic changes in output current, as the control system successfully restored the output voltage close to its nominal value after the transient deviation.

During the project, particular attention was given to maintaining the component selection within reasonable margins to avoid unnecessary oversizing, ensuring a balance between performance and cost. Although most practical challenges were adequately addressed, further refinement the heatsink design with even more detailed calculations could be a potential improvement.

Overall, the project met its goals by combining theoretical analysis, component dimensioning, and simulations, in order to deliver a reliable and efficient buck converter design aligned with the essential specifications.

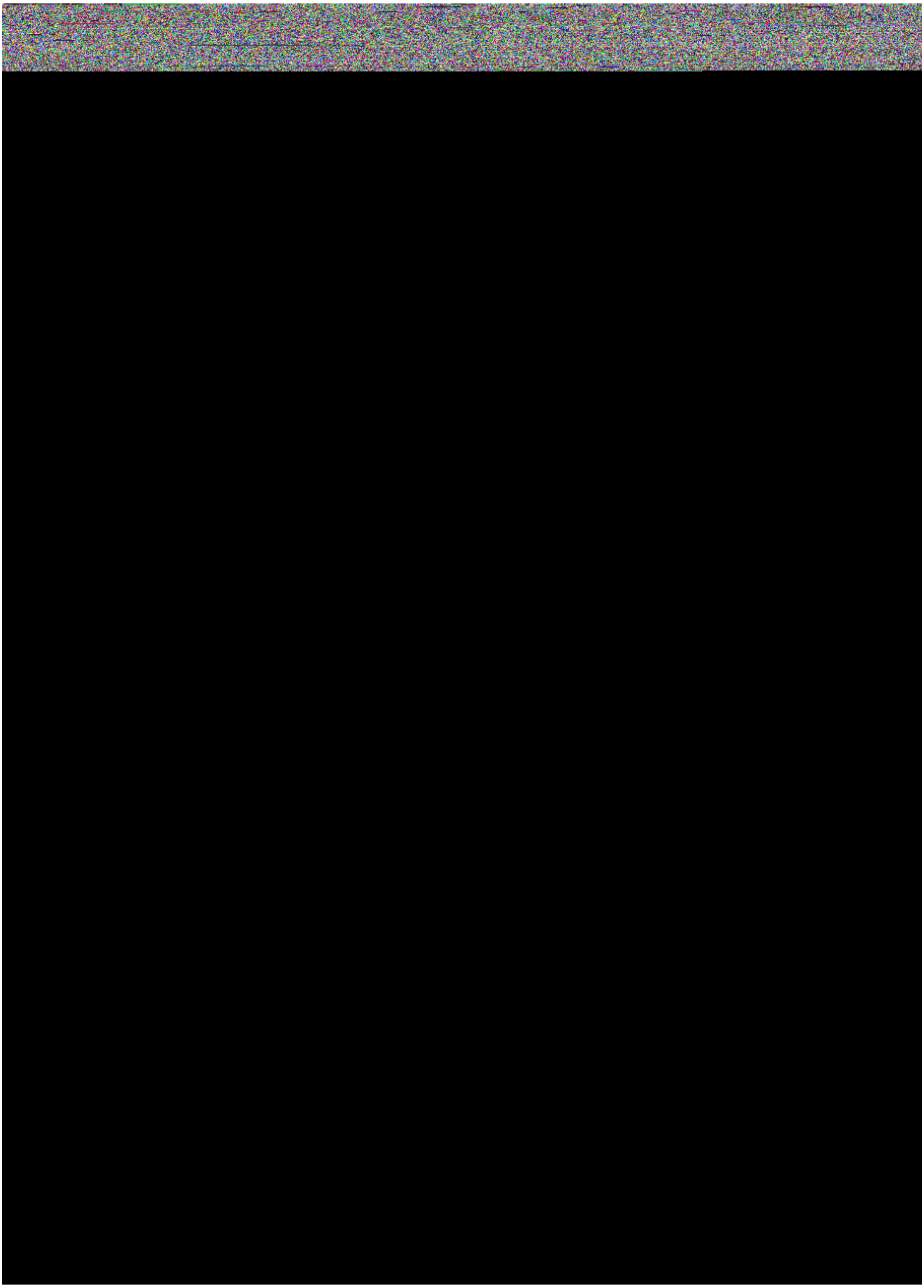
References

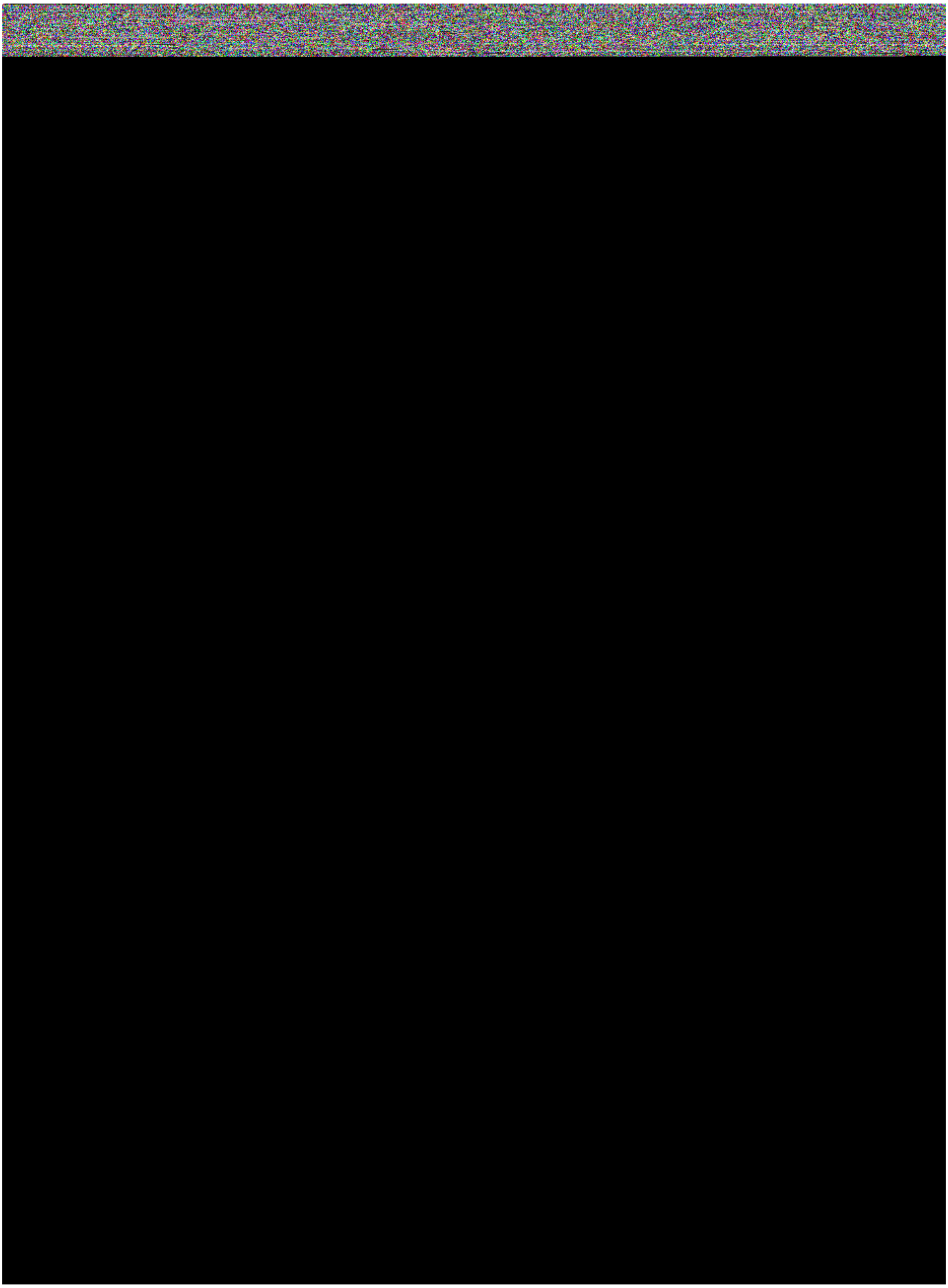
[1] N. Mohan, T. M. Undeland, and W. P. Robbins, *Power Electronics: Converters, Applications, and Design*. John Wiley & Sons, 2003.

Appendix

Parameter	**Value**	**Description**
v_{in}	$500\,\text{V} - 580\,\text{V}$	Input voltage range
V_o	$260\,\text{V}$	Output voltage
Δv_o	$4\,\text{V}$	Voltage ripple
$I_{o,\max}$	$20\,\text{A}$	Max output current
L_σ	$200\,\text{nH}$	Stray inductance
$C \cdot L$	$< 9 \cdot 10^{-8}\, s^2$	Capacitance–inductance limit
r_L	$K_L \cdot L^{0.7}$	DC resistance ($K_L = 7.97$)
r_C	K_{ESR}/C	ESR model ($K_{\text{ESR}} = 8.5\,\mu\text{F}\Omega$)
$T_{j,\max}$	$150°\text{C}$	Max junction temp
$T_{a,\max}$	$50°\text{C}$	Max ambient temp

Table 4: Dataset D





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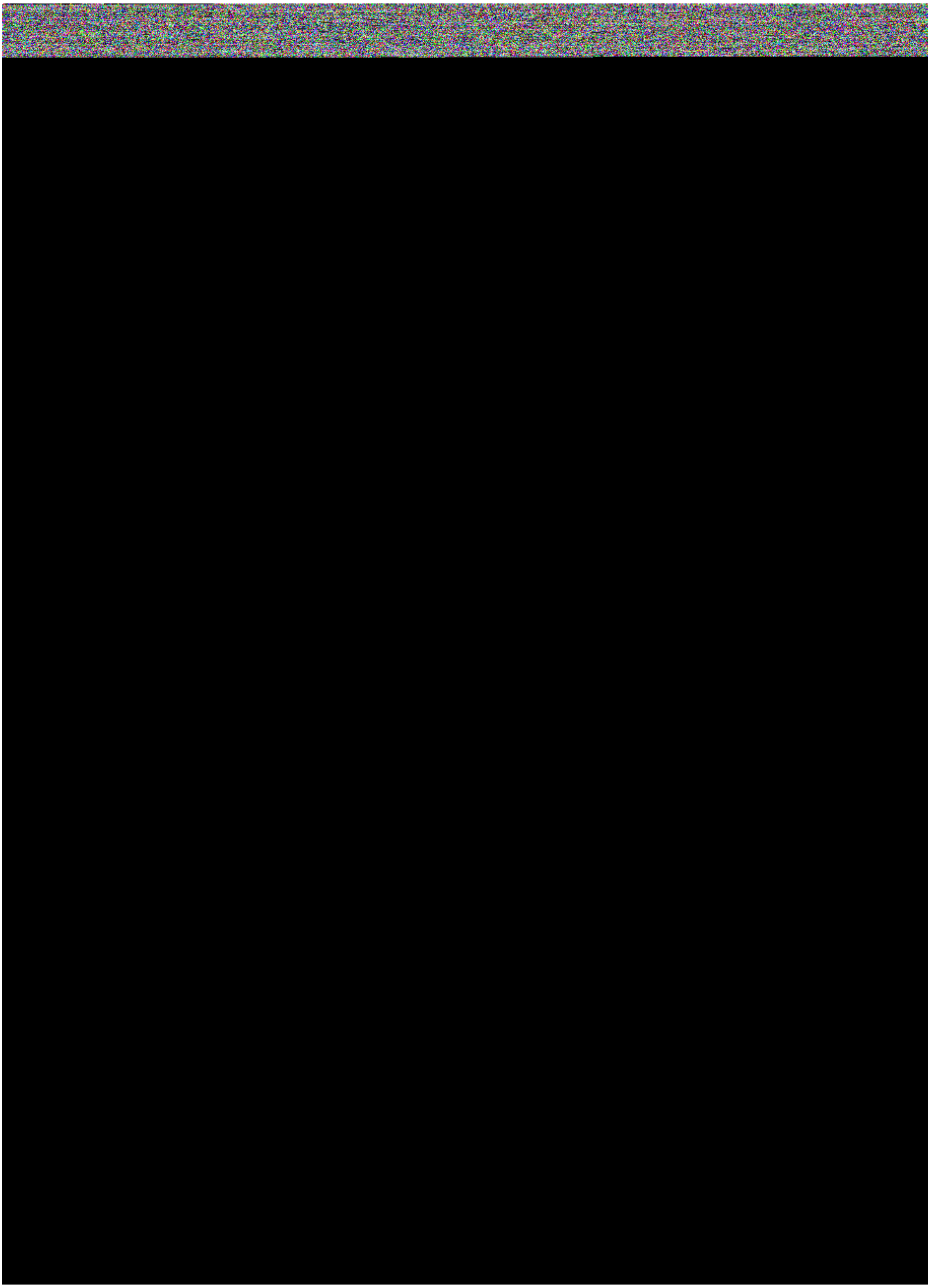

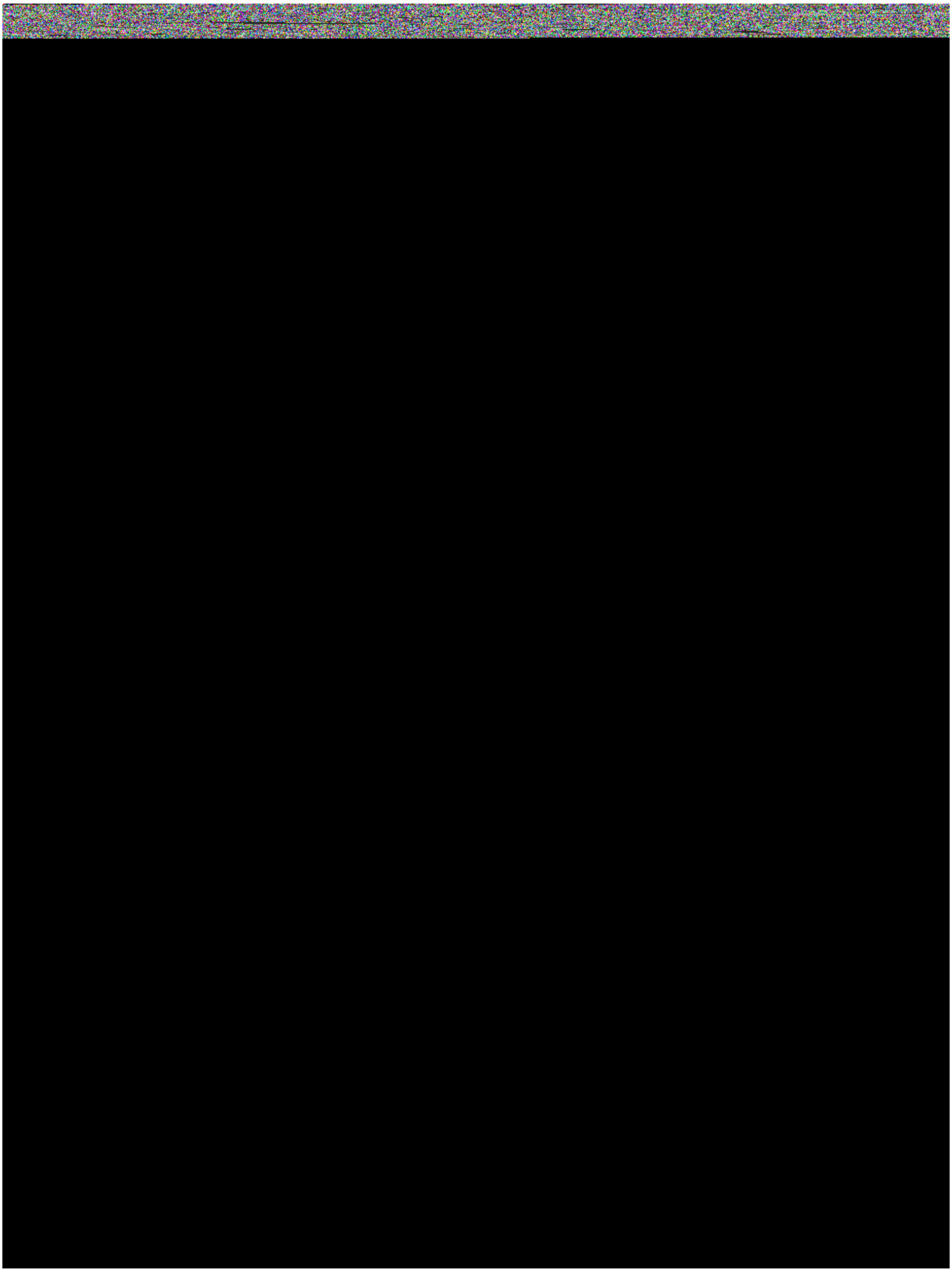

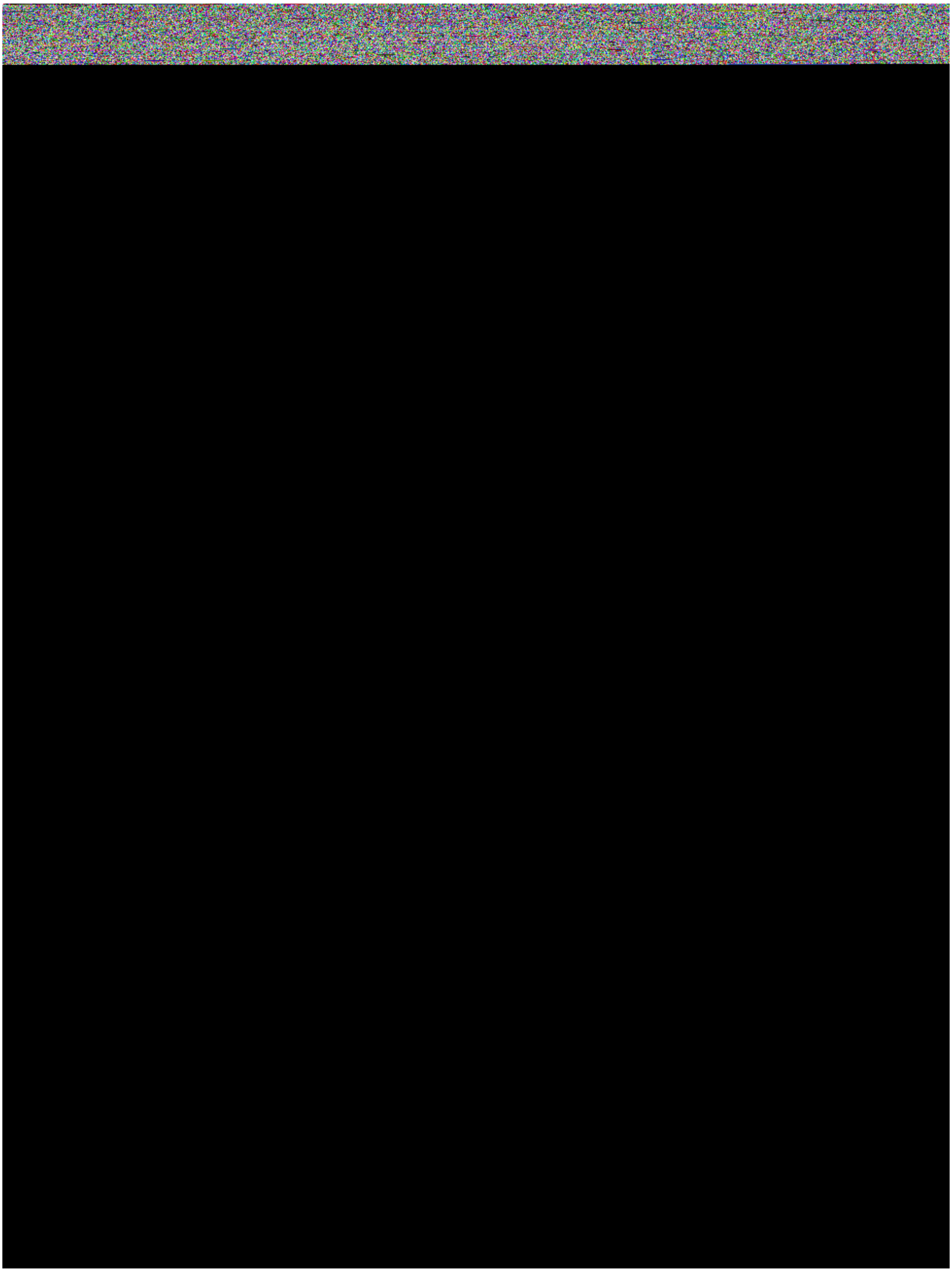

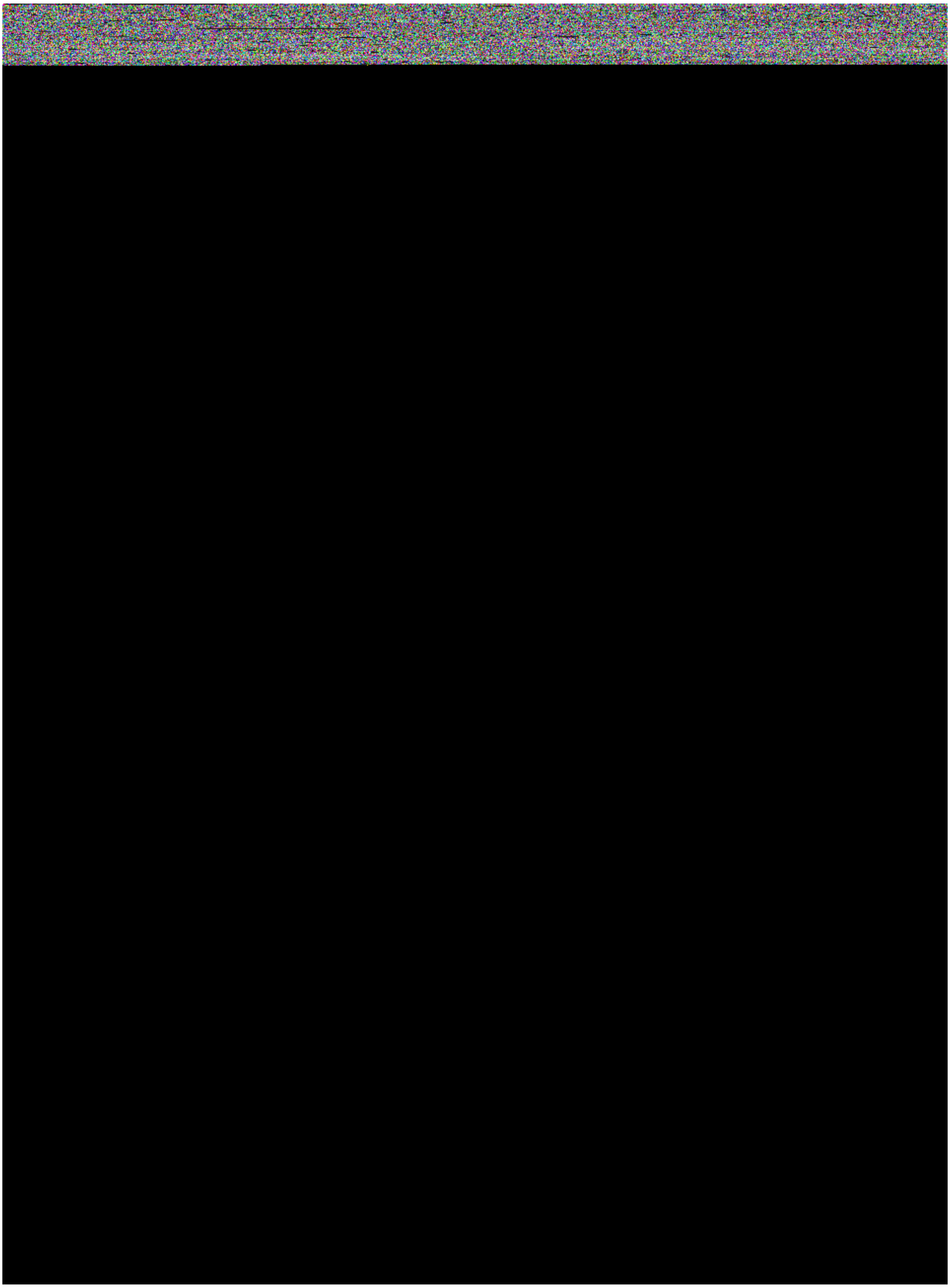

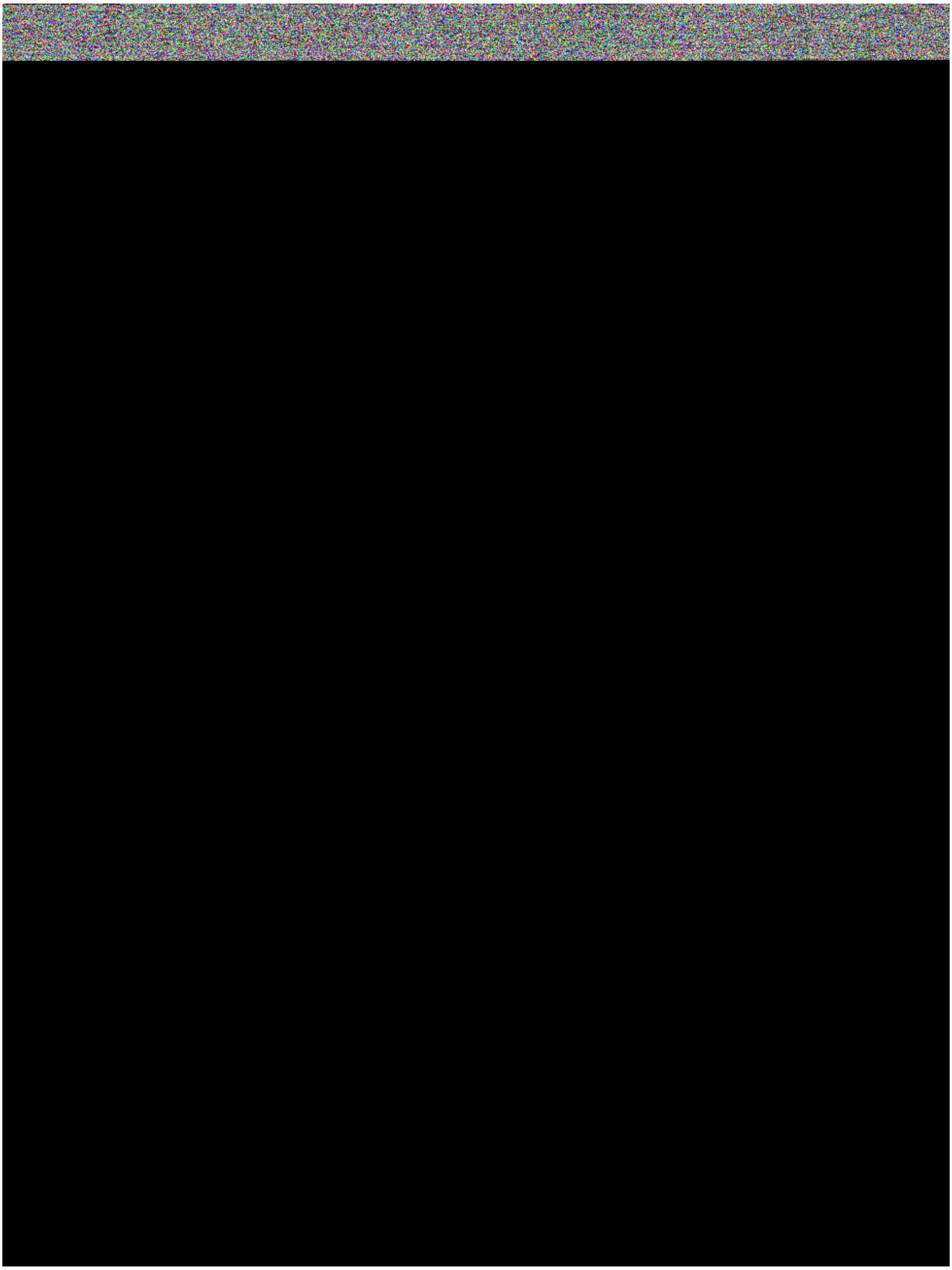

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